

Álvaro Ras Carmona, PhD

AI/ML Scientist · Computational Biology · Madrid, Spain

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Portfolio-generated profile summary based on CV information; not an original institutional CV.

PROFILE

AI/ML scientist with 8 years of experience in biological data analysis, biostatistics and large-scale biological datasets. Classical ML, deep learning and encoder-only transformers applied to biology; end-to-end model preparation, training and evaluation. Current focus: AI for drug repurposing.

RESEARCH EXPERIENCE

Computational Biology & AI Research Scientist / Postdoctoral Researcher · Macromolecular Assemblies Structure Group, CIB Margarita Salas-CSIC, Madrid · Jun 2025–present

- Mixture of Experts composed of multiple MLPs for binary classification; protein embeddings using ESM-2, ProtBERT and ProtT5.
- RAG + LLMs via Ollama to predict clinical success of therapeutic candidates for drug repurposing; in silico peptide candidate design using BindCraft and BinderFlow.

Research project funded in part through the Silicon Valley Community Foundation.

Immunomedicine / Immunology research · Universidad Complutense de Madrid · Jul 2024–Jun 2025 and Oct 2019–Jul 2024

- ML classification of B-cell epitope peptides; t-SNE/UMAP, BLAST, functional enrichment and deployment of predictive models as web tools.

EDUCATION & INTERNATIONAL RESEARCH

- PhD in Biomedical Research (Bioinformatics), Universidad Complutense de Madrid · Oct 2019–Jul 2024 · Cum Laude.
- MSc in Omics Data Analysis · Oct 2018–Sep 2019 · grade 8.1. BSc in Biotechnology · Sep 2014–Jul 2018 · grade 7.8.
- University of Catania, Computational Modeling in Systems Biomedicine · 19 Sep–1 Dec 2022 · Supervisor: Prof. Francesco Pappalardo. Genetic algorithms for prediction of the next SARS-CoV-2 variant.

TECHNICAL SKILLS

AI/ML: Python, R, PyTorch, TensorFlow, scikit-learn, Transformers, Mixture of Experts, MLPs, LLMs, RAG, NLP, genetic algorithms, cross-validation, model evaluation, Optuna.

Biology & data: ESM-2, ProtBERT, ProtT5, BioPython, biostatistics, AlphaFold3, BindCraft, BinderFlow, AlphaPulldown, Scipion-Chem, immunoinformatics, NumPy, Pandas, SQL, UMAP, t-SNE.

Infrastructure: HPC, GPU, Slurm, Docker, Ollama, Linux, Bash, Git/GitHub.

RESEARCH RECORD

16 publications/articles/book chapters (9 as first or co-first author) and 14 conference contributions.

SELECTED PUBLICATIONS

2025 · Di Salvatore et al. — Exploring SARS-CoV-2 spike protein mutations through genetic algorithm-driven structural modeling. *Bioinformatics Advances*. DOI: [10.1093/bioadv/vbaf288](https://doi.org/10.1093/bioadv/vbaf288)

2025 · Ras-Carmona et al. — Similarity to Self-Antigens Shapes Epitope Recognition from Viruses Under Autoimmune and Infectious Disease. *IJMS*. DOI: [10.3390/ijms26136041](https://doi.org/10.3390/ijms26136041)

2024 · Ras-Carmona & Reche — Analysis of Virus-Specific B Cell Epitopes Reveals Extensive Antigen Degradation Prior to Recognition. *Cells*. DOI: [10.3390/cells1313107](https://doi.org/10.3390/cells1313107)

2022 · Ras-Carmona et al. — Prediction of B cell epitopes in proteins using a novel sequence similarity-based method. *Scientific Reports*. DOI: [10.1038/s41598-022-18021-1](https://doi.org/10.1038/s41598-022-18021-1)

2021 · Ras-Carmona et al. — BCEPS: A Web Server to Predict Linear B Cell Epitopes with Enhanced Immunogenicity and Cross-Reactivity. *Cells*. DOI: [10.3390/cells10102744](https://doi.org/10.3390/cells10102744)

LATEST TRAINING

AIHUB Summer School for Early-Career Researchers in AI Applied to Science · Sep 2026. GPU-Based Parallel Supercomputing Workshop at HCP DRAGO · May 2026.